Containerization of Serverless Workloads and Understanding Its Cost

Aaryan Bhagat : 862468325
Soorya Saravannan : 862289961

Goal

Serverless computing offers scalability and cost efficiency but faces challenges like cold starts and vendor lock-in. Containerization provides portability and flexibility. This project compares the cost and performance of both approaches to identify their strengths and best-use scenarios.

Design

- Kubernetes Cluster consisting of multiple nodes and a master node.
- Each node will have multiple pods which will run either only docker containers or knative serverless applications.
- The nodes will be maximally utilized using vertical scaling.

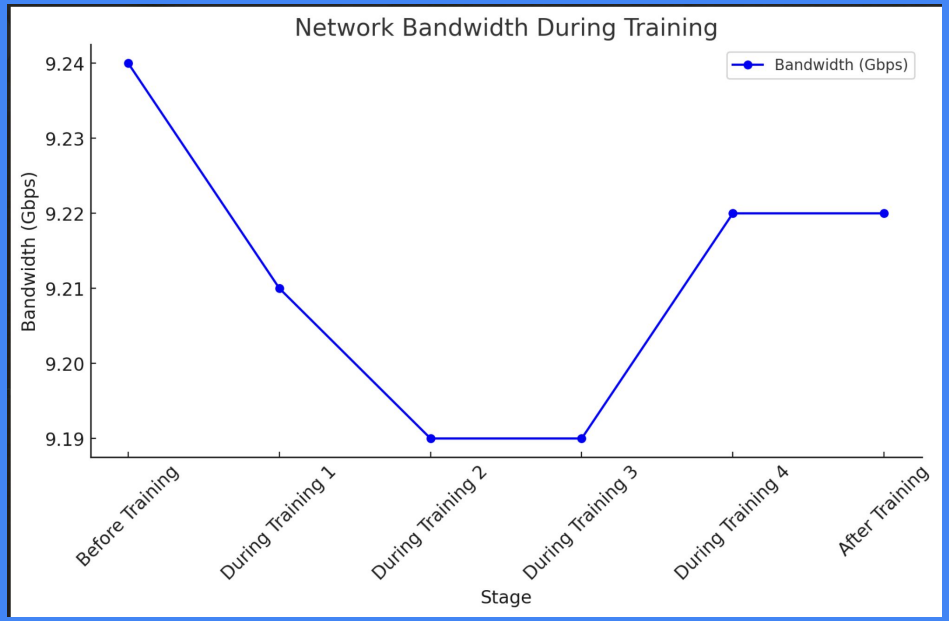
Implementation

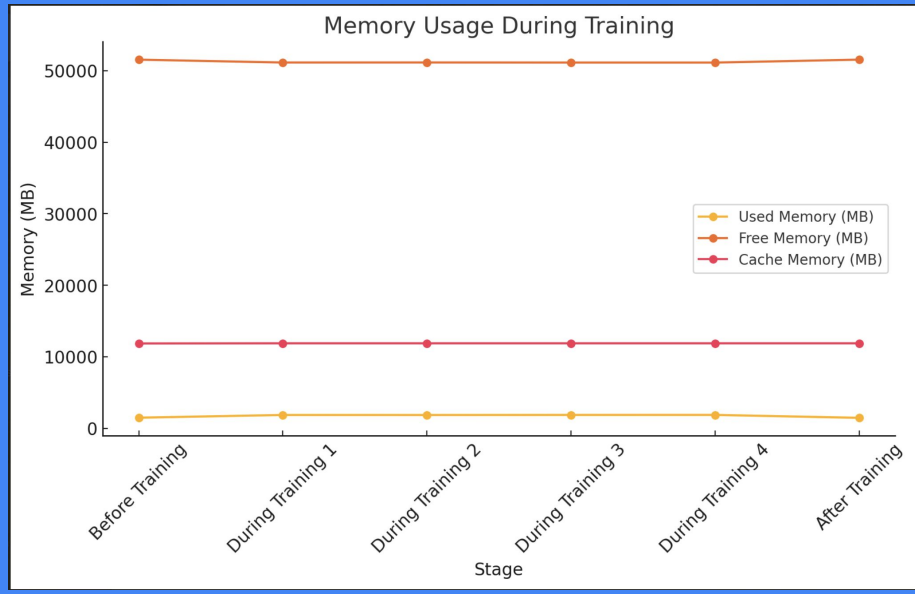
- Creating a bare metal kubernetes cluster on cloud lab.
- Since serverless and containers have their own pros and cons, different tasks were made to test each one of their strengths and weaknesses.
- Inference code is packaged into a Docker container with all dependencies.
- Three Docker containers are manually run to simulate container level execution.
- Inference container is deployed as a Knative service on Kubernetes.
- Inference script logs CPU, memory, and network metrics during execution
- Comparisons were made between the two approaches

Demo

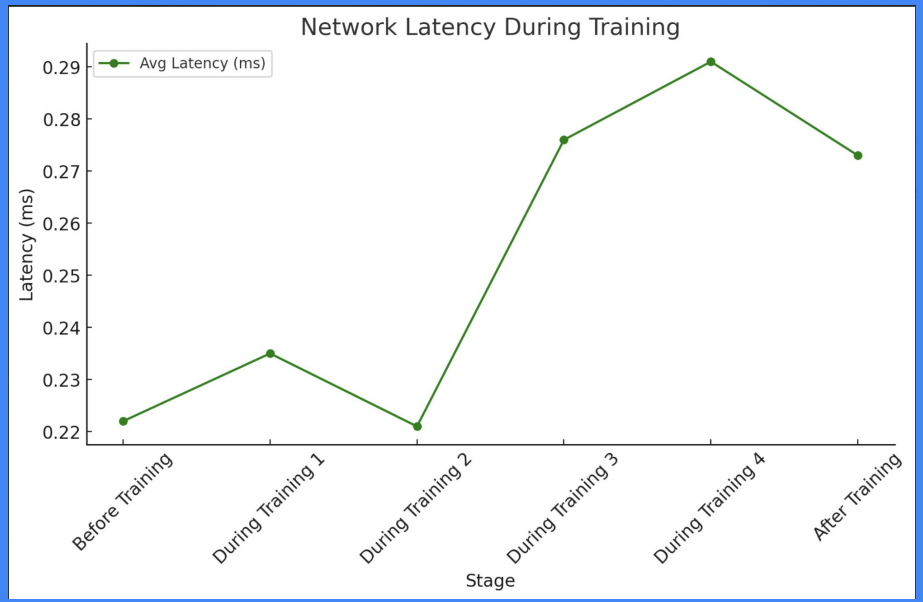
Load testing - serverless
Docker
Knative
Codebase

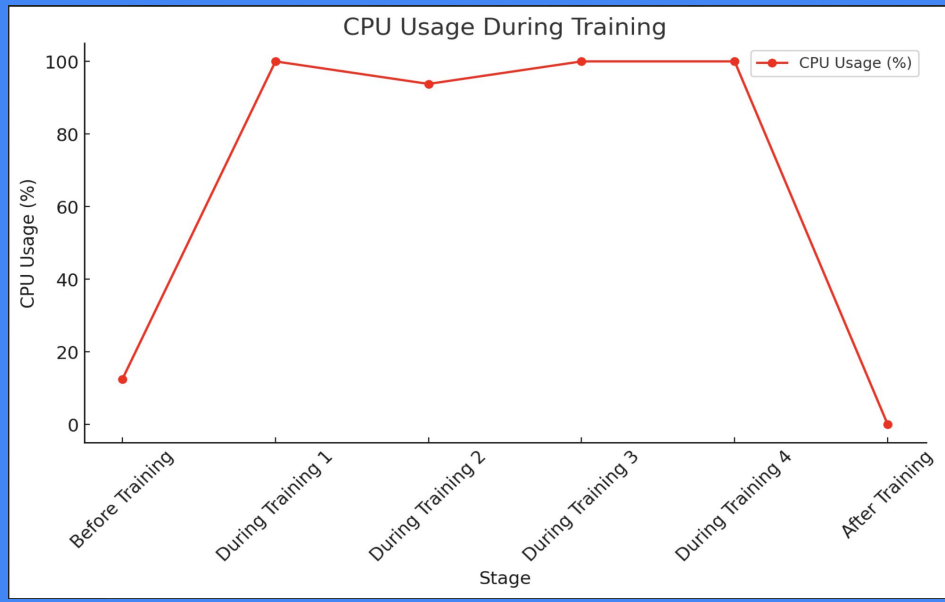
Some results



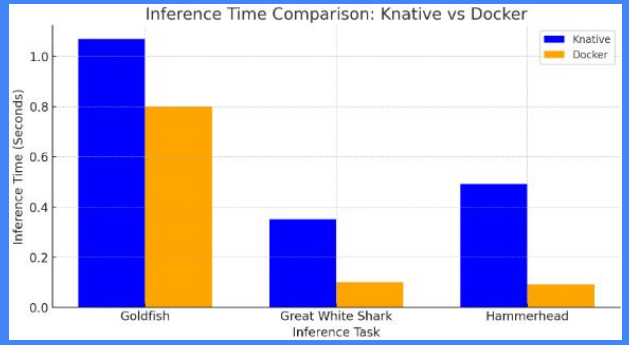


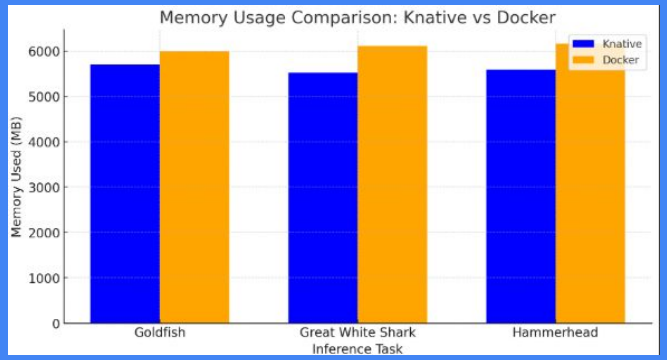
Some results

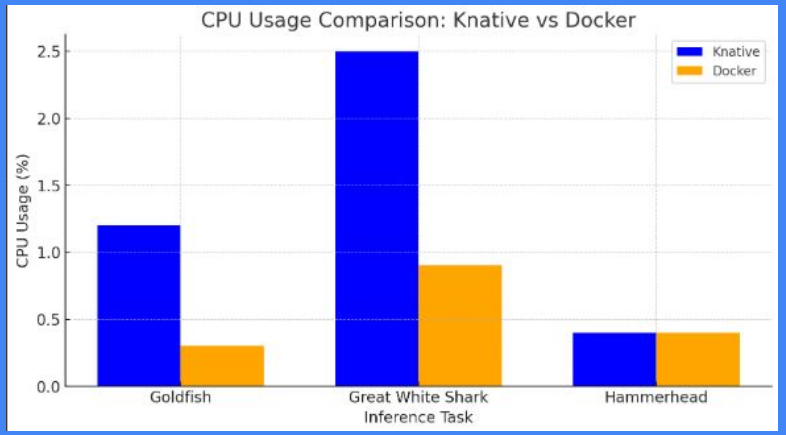




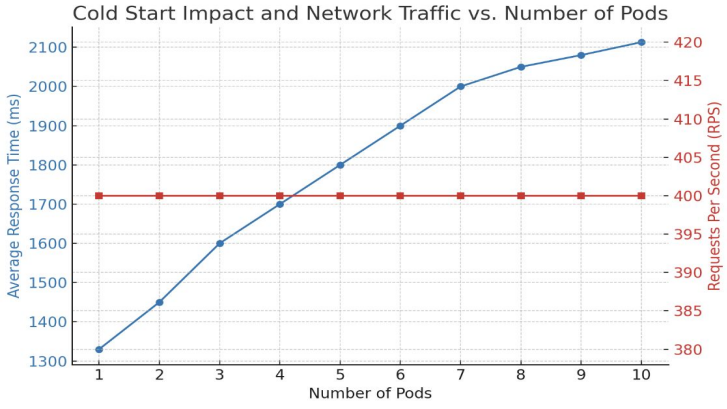
Some results







Results of Autoscaling Serverless



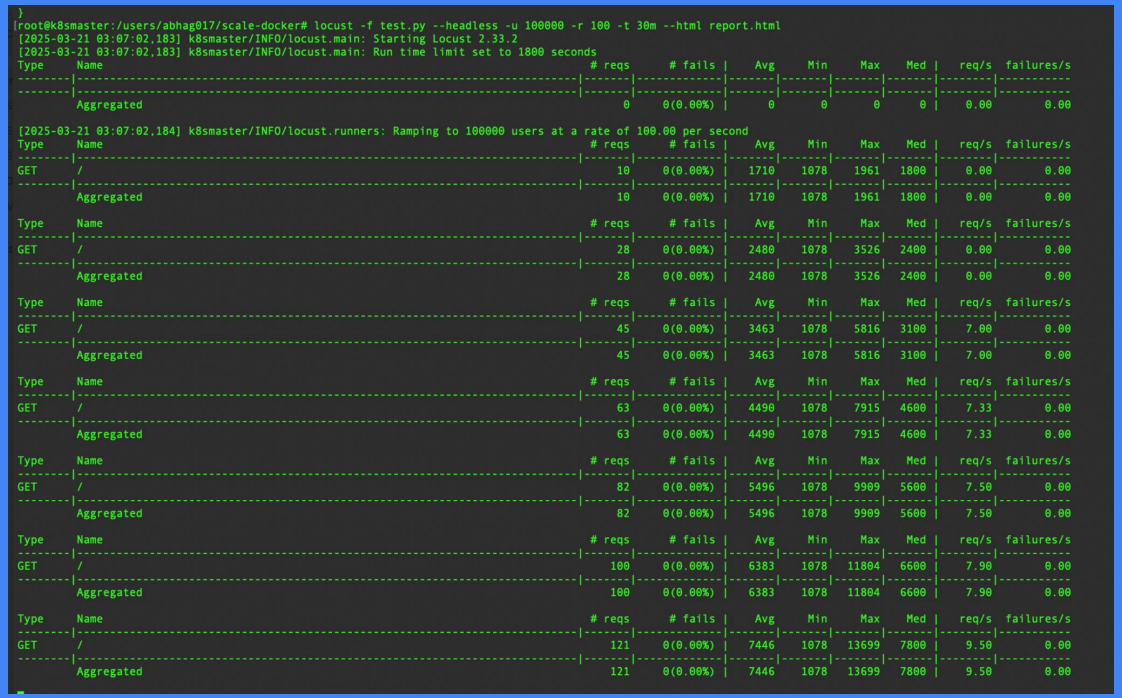

```
Non-terminated Pods:         (7 in total)
  Namespace                  Name                                    CPU Requests  CPU Limits  Memory Requests  Memory Limits  Age
  ---------                  ----                                    ------------  ----------  ---------------  -------------  ---
  knative-serving            net-kourier-controller-75774fd8d5-mllnc 200m (0%)     500m (0%)   200Mi (0%)       500Mi (0%)     7h38m
  kourier-system             3scale-kourier-gateway-58c8cb46c6-2l5jl 200m (0%)     500m (0%)   200Mi (0%)       500Mi (0%)     7h38m
  kube-system                calico-node-r9fkt                       250m (0%)     0 (0%)      0 (0%)           0 (0%)         8h
  kube-system                kube-proxy-h5jtq                        0 (0%)        0 (0%)      0 (0%)           0 (0%)         8h
  kube-system                metrics-server-75bf97fcc9-7fz9w         100m (0%)     0 (0%)      200Mi (0%)       0 (0%)         162m
  metallb-system             controller-5b5b575d55-z6wnd             0 (0%)        0 (0%)      0 (0%)           0 (0%)         150m
  metallb-system             speaker-l9w4r                           0 (0%)        0 (0%)      0 (0%)           0 (0%)         150m
Allocated resources:
```

Sending request to 1 flask container

```
[2025-03-21 03:02:25,690] k8smaster/INFO/locust.runners: Ramping to 100000 users at a rate of 100.00 per second
Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|-----------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /                                                                                4     0(0.00%) |   1353    1036    1827   1200 |    0.00        0.00
--------|-----------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                       4     0(0.00%) |   1353    1036    1827   1200 |    0.00        0.00

Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|-----------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /                                                                                9     0(0.00%) |   1880    1036    2822   2000 |    0.00        0.00
--------|-----------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                       9     0(0.00%) |   1880    1036    2822   2000 |    0.00        0.00

Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|-----------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /                                                                               16     0(0.00%) |   3256    1036    5907   2300 |    2.25        0.00
--------|-----------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                      16     0(0.00%) |   3256    1036    5907   2300 |    2.25        0.00

Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|-----------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /                                                                               20     0(0.00%) |   4001    1036    7805   4300 |    2.33        0.00
--------|-----------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                      20     0(0.00%) |   4001    1036    7805   4300 |    2.33        0.00

Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|-----------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /                                                                               25     0(0.00%) |   4925    1036    9095   4800 |    2.43        0.00
--------|-----------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                      25     0(0.00%) |   4925    1036    9095   4800 |    2.43        0.00

Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|-----------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /                                                                               31     0(0.00%) |   6137    1036   11918   5900 |    2.50        0.00
--------|-----------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                      31     0(0.00%) |   6137    1036   11918   5900 |    2.50        0.00

Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|-----------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /                                                                               36     0(0.00%) |   7051    1036   13715   6400 |    2.60        0.00
--------|-----------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                      36     0(0.00%) |   7051    1036   13715   6400 |    2.60        0.00

Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|-----------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /                                                                               41     0(0.00%) |   8003    1036   15416   8000 |    2.60        0.00
--------|-----------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                      41     0(0.00%) |   8003    1036   15416   8000 |    2.60        0.00

Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|-----------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /                                                                               48     0(0.00%) |   9273    1036   17512   9100 |    2.60        0.00
--------|-----------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                      48     0(0.00%) |   9273    1036   17512   9100 |    2.60        0.00
```

```
Every 2.0s: kubectl describe node k8sworker1.example.net | grep -B 20 -A 200 Requests

  cpu:                128
  ephemeral-storage:  60345081347
  hugepages-1Gi:      0
  hugepages-2Mi:      0
  memory:             263451944Ki
  pods:               110
System Info:
  Machine ID:                 7c52626e573b484f813fed3055eb9193
  System UUID:                4c4c4544-0033-3510-8034-b1c04f324733
  Boot ID:                    8e5f57fa-e0eb-4784-898f-28edf4ebf7a5
  Kernel Version:             5.15.0-131-generic
  OS Image:                   Ubuntu 22.04.2 LTS
  Operating System:           linux
  Architecture:               amd64
  Container Runtime Version:  containerd://1.7.24
  Kubelet Version:            v1.28.15
  Kube-Proxy Version:         v1.28.15
PodCIDR:                      192.168.2.0/24
PodCIDRs:                     192.168.2.0/24
Non-terminated Pods:          (8 in total)
  Namespace                   Name                                                  CPU Requests  CPU Limits  Memory Requests  Memory Limits  Age
  ---------                   ----                                                  ------------  ----------  ---------------  -------------  ---
  default                     flask-random-file-generator-7f68d5b9f6-rgjhz          1m (0%)       3m (0%)     32Mi (0%)        64Mi (0%)      6m37s
  default                     ml-inference-docker-6c6978ddc4-ncq4x                  0 (0%)        0 (0%)      0 (0%)           0 (0%)         6h24m
  keda                        keda-admission-7dc88f744c-tnn64                       100m (0%)     1 (0%)      100Mi (0%)       1000Mi (0%)    132m
  keda                        keda-metrics-apiserver-758dc4f488-lftvw               100m (0%)     1 (0%)      100Mi (0%)       1000Mi (0%)    132m
  keda                        keda-operator-85bc6c668d-ddhg9                        100m (0%)     1 (0%)      100Mi (0%)       1000Mi (0%)    132m
  kube-system                 calico-node-crggz                                     250m (0%)     0 (0%)      0 (0%)           0 (0%)         6h47m
  kube-system                 kube-proxy-rtmzt                                      0 (0%)        0 (0%)      0 (0%)           0 (0%)         6h51m
  kube-system                 metrics-server-75bf97fcc9-47tq6                       100m (0%)     0 (0%)      200Mi (0%)       0 (0%)         6h44m
Allocated resources:
  (Total limits may be over 100 percent, i.e., overcommitted.)
  Resource           Requests     Limits
  --------           --------     ------
  cpu                651m (0%)    3003m (2%)
  memory             532Mi (0%)   3064Mi (1%)
  ephemeral-storage  0 (0%)       0 (0%)
  hugepages-1Gi      0 (0%)       0 (0%)
  hugepages-2Mi      0 (0%)       0 (0%)
Events:              <none>
```

Sending request to 3 flask containers



```
}
[root@k8smaster:/users/abhag017/scale-docker# locust -f test.py --headless -u 100000 -r 100 -t 30m --html report.html
[2025-03-21 03:07:02,183] k8smaster/INFO/locust.main: Starting Locust 2.33.2
[2025-03-21 03:07:02,183] k8smaster/INFO/locust.main: Run time limit set to 1800 seconds
Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|-----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                       0     0(0.00%) |      0       0       0      0 |    0.00        0.00

[2025-03-21 03:07:02,184] k8smaster/INFO/locust.runners: Ramping to 100000 users at a rate of 100.00 per second
Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|-----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /                                                                               10     0(0.00%) |   1710    1078    1961   1800 |    0.00        0.00
--------|-----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                      10     0(0.00%) |   1710    1078    1961   1800 |    0.00        0.00

Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|-----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /                                                                               28     0(0.00%) |   2480    1078    3526   2400 |    0.00        0.00
--------|-----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                      28     0(0.00%) |   2480    1078    3526   2400 |    0.00        0.00

Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|-----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /                                                                               45     0(0.00%) |   3463    1078    5816   3100 |    7.00        0.00
--------|-----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                      45     0(0.00%) |   3463    1078    5816   3100 |    7.00        0.00

Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|-----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /                                                                               63     0(0.00%) |   4490    1078    7915   4600 |    7.33        0.00
--------|-----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                      63     0(0.00%) |   4490    1078    7915   4600 |    7.33        0.00

Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|-----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /                                                                               82     0(0.00%) |   5496    1078    9909   5600 |    7.50        0.00
--------|-----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                      82     0(0.00%) |   5496    1078    9909   5600 |    7.50        0.00

Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|-----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /                                                                              100     0(0.00%) |   6383    1078   11804   6600 |    7.90        0.00
--------|-----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                     100     0(0.00%) |   6383    1078   11804   6600 |    7.90        0.00

Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|-----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /                                                                              121     0(0.00%) |   7446    1078   13699   7800 |    9.50        0.00
--------|-----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                     121     0(0.00%) |   7446    1078   13699   7800 |    9.50        0.00
```

When requests are increased 2 containers are created in one node

```
Every 2.0s: kubectl describe node k8sworker1.example.net | grep -B 20 -A 200 Requests

  cpu:                128
  ephemeral-storage:  60345081347
  hugepages-1Gi:      0
  hugepages-2Mi:      0
  memory:             263451944Ki
  pods:               110
System Info:
  Machine ID:                 7c52626e573b484f813fed3055eb9193
  System UUID:                4c4c4544-0033-3510-8034-b1c04f324733
  Boot ID:                    8e5f57fa-e0eb-4784-898f-28edf4ebf7a5
  Kernel Version:             5.15.0-131-generic
  OS Image:                   Ubuntu 22.04.2 LTS
  Operating System:           linux
  Architecture:               amd64
  Container Runtime Version:  containerd://1.7.24
  Kubelet Version:            v1.28.15
  Kube-Proxy Version:         v1.28.15
PodCIDR:                      192.168.2.0/24
PodCIDRs:                     192.168.2.0/24
Non-terminated Pods:          (9 in total)
  Namespace                   Name                                                  CPU Requests  CPU Limits  Memory Requests  Memory Limits  Age
  ---------                   ----                                                  ------------  ----------  ---------------  -------------  ---
  default                     flask-random-file-generator-688d9fbcfb-62rmg          1m (0%)       3m (0%)     32Mi (0%)       64Mi (0%)      2m56s
  default                     flask-random-file-generator-688d9fbcfb-mzk6k          1m (0%)       3m (0%)     32Mi (0%)       64Mi (0%)      2m47s
  default                     ml-inference-docker-6c6978ddc4-ncq4x                  0 (0%)        0 (0%)      0 (0%)          0 (0%)         6h29m
  keda                        keda-admission-7dc88f744c-tnn64                       100m (0%)     1 (0%)      100Mi (0%)      1000Mi (0%)    137m
  keda                        keda-metrics-apiserver-758dc4f488-lftvw               100m (0%)     1 (0%)      100Mi (0%)      1000Mi (0%)    137m
  keda                        keda-operator-85bc6c668d-ddhg9                        100m (0%)     1 (0%)      100Mi (0%)      1000Mi (0%)    137m
  kube-system                 calico-node-crggz                                     250m (0%)     0 (0%)      0 (0%)          0 (0%)         6h51m
  kube-system                 kube-proxy-rtmzt                                      0 (0%)        0 (0%)      0 (0%)          0 (0%)         6h55m
  kube-system                 metrics-server-75bf97fcc9-47tq6                       100m (0%)     0 (0%)      200Mi (0%)      0 (0%)         6h48m
Allocated resources:
  (Total limits may be over 100 percent, i.e., overcommitted.)
  Resource           Requests    Limits
  --------           --------    ------
  cpu                652m (0%)   3006m (2%)
  memory             564Mi (0%)  3128Mi (1%)
  ephemeral-storage  0 (0%)      0 (0%)
  hugepages-1Gi      0 (0%)      0 (0%)
  hugepages-2Mi      0 (0%)      0 (0%)
Events:              <none>
```

one is created in another node

```
Every 2.0s: kubectl describe node k8sworker2.example.net | grep -B 20 -A 200 Requests

  cpu:                40
  ephemeral-storage:  60345081347
  hugepages-1Gi:      0
  hugepages-2Mi:      0
  memory:             263926772Ki
  pods:               110
System Info:
  Machine ID:                 46967b30eae142f8af3fc301cb812b7d
  System UUID:                4c4c4544-0031-4610-804a-c4c04f383432
  Boot ID:                    54e59b62-cbcc-4158-a155-788aadcc7337
  Kernel Version:             5.15.0-131-generic
  OS Image:                   Ubuntu 22.04.2 LTS
  Operating System:           linux
  Architecture:               amd64
  Container Runtime Version:  containerd://1.7.24
  Kubelet Version:            v1.28.15
  Kube-Proxy Version:         v1.28.15
PodCIDR:                      192.168.1.0/24
PodCIDRs:                     192.168.1.0/24
Non-terminated Pods:          (4 in total)
  Namespace                   Name                                          CPU Requests  CPU Limits  Memory Requests  Memory Limits  Age
  ---------                   ----                                          ------------  ----------  ---------------  -------------  ---
  default                     flask-random-file-generator-688d9fbcfb-2hnlm  1m (0%)       3m (0%)     32Mi (0%)       64Mi (0%)      3m15s
  default                     python-http-server-598cb4b7d7-95fg5           10m (0%)      50m (0%)    32Mi (0%)       64Mi (0%)      80m
  kube-system                 calico-node-rj2ct                             250m (0%)     0 (0%)      0 (0%)          0 (0%)         6h52m
  kube-system                 kube-proxy-w9rw9                              0 (0%)        0 (0%)      0 (0%)          0 (0%)         6h56m
Allocated resources:
  (Total limits may be over 100 percent, i.e., overcommitted.)
  Resource           Requests   Limits
  --------           --------   ------
  cpu                261m (0%)  53m (0%)
  memory             64Mi (0%)  128Mi (0%)
  ephemeral-storage  0 (0%)     0 (0%)
  hugepages-1Gi      0 (0%)     0 (0%)
  hugepages-2Mi      0 (0%)     0 (0%)
Events:              <none>
```

Learning from errors

- Kubernetes autoscaling, specifically the part which required to create more cloudlab nodes needed an internal integration with the cloudlab environment.
- Sadly this integration was not present, the alternate option was to create ample amount of nodes and give a custom load balancer however this will assume all the run times for the containers are already up and running so it will not intake the startup times of the containers.
- Load tests need to be efficiently designed to stress test docker containers as they have good memory and cpu cycles associated with it.

Eventual Realization

- Event based tasks serverless still holds superiority both in terms of cost and speed (inference for e.g).
- Containers can still undergo tasks having high amount of load and it needs less wrappers to check the metrics.
- Cost analysis for both of these computing beds are different
- Its easy to vertically scale a serverless than a container (our experimental realization).

Thanks!

Questions and Answers